Remote Sensing Image Scene Classification via Graph Template Enhancement and Supplementation Network With Dual-Teacher Knowledge Distillation

Wujie Zhou, *Senior Member, IEEE*, Penghan Yang, Yuanyuan Liu, *Member, IEEE*, Runmin Cong, *Senior Member, IEEE*, and Qiuping Jiang, *Senior Member, IEEE*

***Abstract*— Image-processing techniques used for remote sensing images (RSIs) have addressed the issues associated with scene classification. However, two challenges remain; first, most bi-modal methods integrate the extracted features indiscriminately while neglecting certain more relevant features, and bi-modal feature qualities may vary in different scenarios. Second, previous studies have improved the feature extraction performance at the expense of computational speed and complexity, hindering the wide application of improved versions. To address these issues, we constructed a graph template enhancement and supplementation network (GTESNet) with dual-teacher knowledge distillation (KD), GTESNet-S*, to emphasize and manage the extracted features adaptively and compress the model. First, we developed a graph template enhancement (GTE) module that combined long-range contextual information with clustered template features. Second, an exchange-correlation fusion (ECF) module was introduced to allocate feature distribution dynamically and achieve integration. Third, we constructed feedback complementary decoders (FCDs) consisting of two subdecoders with a cascade connection that used a feedback mechanism to supplement the final output. A dual-teacher distillation method that utilized simple confidence perceptions to assign weights to both teachers was implemented during KD. Furthermore, we introduced clustered templates and graph-relation distillations (GRDs), allowing the student GTESNet-S to learn the clustering ability and semantic association process of the teacher GTESNet-T. Finally, we introduced simple progressive decoder distillation (PDD) to facilitate the learning of GTESNet-S. Extensive experiments on two publicly available datasets demonstrated that the GTESNet outperformed similar state-of-the-art (SOTA) methods. The code is available at: https://github.com/MAXHAN22/GTESNet.**

***Index Terms*— Clustered template, dual-teacher knowledge distillation (KD), graph template, scene classification.**

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Wujie Zhou and Penghan Yang are with the School of Information and Electronic Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, China (e-mail: wujiezhou@163.com).

Yuanyuan Liu is with the School of Computer Science and Engineering, Nanyang Technological University, Singapore 308232, and also with the School of Computer and Technology, China University of Geosciences, Wuhan 430074, China (e-mail: liuyy@cug.edu.cn).

Runmin Cong is with the School of Control Science and Engineering, Shandong University, Jinan 250100, China (e-mail: rmcong@sdu.edu.cn).

Qiuping Jiang is with the School of Information Science and Engineering, Ningbo University, Ningbo 315211, China (e-mail: jiangqiuping@nbu.edu.cn).

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I. Introduction

SATELLITES equipped with high-resolution image acquisition have been used increasingly in recent years, resulting in numerous remote sensing images (RSIs). RSI scene classification labels high-resolution RSIs correctly using predefined semantic categories based on their semantic differences. This process helps in understanding and analyzing the rich information contained in the images, thereby facilitating tasks such as urban planning [1], environmental protection [2], and land-cover mapping [3].

Although scene classification has yielded promising results in recent years of focused research and continuous technological advancements [4], [5], [6], [7], unimodal approaches continue to face challenges when processing complex scenes where the objects are visually similar to the background or exhibit low contrast. Therefore, recent studies [8], [9], [10], [11], [12] have favored the introduction of data from another modality as an auxiliary that provides additional information. Properly utilizing the complementarity of multimodal data can improve the performance of computer vision systems for complex scenes. Herein, we introduced the normalized digital surface model (nDSM) as auxiliary data for infrared-red-green (IRRG) images. IRRG offers sufficient color information, whereas nDSM supplies abundant height information that enhances the model's understanding of the scene. However, merging supplemental information may pose new challenges, which are discussed in Section II.

In complex scenes, certain features can easily be misclassified owing to the influence of similar features. Although the aforementioned methods primarily focus on feature fusion, they do not emphasize such features in advance. It is important to utilize long-range contextual information to achieve pixel-level predictions. These tasks have been implemented by executing dilated convolution [13] or passing the feature maps that are learned for each spatial location to their neighbors via the attention mechanism [14], [15]. However, the former approach generates feature representations that are dominated by large targets in the image and performs poorly on small objects, whereas the long-generated feature maps of the latter method have restrictive memory requirements and are not suitable for high-resolution RSIs. Therefore, the distinctiveness of each feature category must be enhanced before the feature fusion step to mine

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the long-range contextual information of large and small targets.

Existing approaches use different fusion strategies to exploit multimodal features [16], [17], [18], [19], [20], [21], [22]. For example, Li et al. [23] applied the attention mechanism across two complementary branches to fuse the low-level detail with high-level semantic features. Morelli et al. [24] presented a detailed comparative analysis of fusion methods applied to RSIs for each of the early, intermediate, and late stages. However, the information contained in multimodal images varies across different scenes, as does their contribution to the interactions in which they are involved. Without applying differentiated operations to these images, these methods interact with the multimodal features directly, resulting in features that do not provide satisfactory results for complex environments. The models used require a large computational capacity and complex structures. Therefore, establishing an accurate and compact model for RSI scene classification is desirable.

Although incorporating auxiliary images enhances the performance of the model when dealing with complex scenes, it does not guarantee the completeness of the prediction. Features at different scales are distinctive, and poorly performing features will interfere with the prediction if multiscale features are directly aggregated without additional manipulation. Previous approaches have aggregated multiscale features by integrating (adding or joining) them layer by layer. Although the aggregation operation is convenient and effective, it neglects the aforementioned problem. Therefore, a more suitable decoder is required to generate the prediction map completely.

This study addresses the first problem by combining a lightweight unsupervised clustering algorithm, clustering template module (CTM), and coordinate graph inference (CGI) module as a graph template enhancement (GTE) module. As shown in Fig. 1, after extracting the features, all expressive features in the RSIs were clustered into template features to capture common intrinsic feature information, enabling our network to focus on features before the fusion period, thereby differentiating between features and reducing the influence of similar features on the model. Following clustering, our CGI module was used to mine long-range contextual information directly (compared with most models that perform graph operations after feature fusion). We propose an exchange-correlation fusion (ECF) module to balance the contributions of the two modalities and completely integrate them. When the linked modal contribution adjustment (MCA) module of ECF adjusts the weights of the modalities adaptively, the correlation matrices of the two modalities are exchanged separately to achieve complete fusion. In addition, we propose feedback complementary decoders (FCDs); each consisting of two cascaded subdecoders, FCD low (FCDL) and FCD high (FCDH) to improve the scene classification quality. An adaptive filtering method was used to restrain superfluous information, avert contamination, and minimize the discrepancies between multiscale features. The predictions of the first subdecoder were employed as supervised inputs into the second subdecoder to complement and correct its outputs and enhance the model performance.

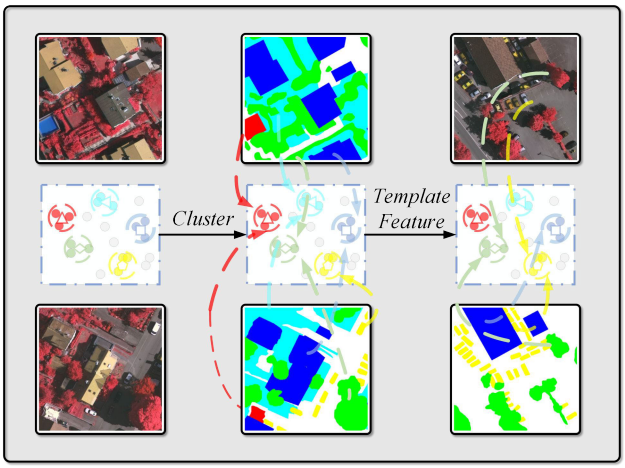


Fig. 1. Template features are learned from all expressive features in the scene, and the common feature information of the objects is clustered. Different colored markers represent different templates, and each template represents the most distinctive feature of a class of objects (e.g., a blue square represents some features of a building).

For the knowledge distillation (KD) period, we designed a new KD strategy that effectively improved the robustness of distilled student models through weighted adaptive collaborative dual-teacher distillation. The KD decision consisted of three aspects: clustered template distillation (CTD), graph-relation distillation (GRD), and progressive decoder distillation (PDD). The CTD allowed GTESNet-S to mimic the ability of GTESNet-T to cluster all expressive features in different scenarios and improve the coding ability of GTESNet-S. The GRD was designed to exploit the full potential of GTESNet-S to mine long-range contextual information and strengthen semantic associations. In addition, we introduced the simple PDD to improve the classification performance of the students. We performed experiments using two public datasets to demonstrate the versatility of GTESNet-S*.

The contributions of this study are summarized as follows.

1) We establish GTE, which uses a clustering algorithm and CTM to generate template features and further differentiate among various features. In addition, we directly use CGI to mine long-range contextual information before proceeding to feature fusion.
2) The MCA addresses the quality imbalance between inputs under different scenarios and the closely connected ECF exchange-correlation matrices from two modal features achieve modal fusion.
3) Our FCDs employ an adaptive filtering method to restrain superfluous information. The prediction result of the first subdecoder is used as feedback to complement and correct the second subdecoder, thereby improving the model performance.
4) We introduce CTD, GRD, and PDD, conveying the knowledge of the respective strengths of the two teachers. Our evaluation using two publicly available datasets showed that the lightweight GTESNet-S* outperformed current state-of-the-art (SOTA) approaches.

The remainder of this article is organized as follows. Section II discusses previous studies on scene classification methods for RSIs. Section III explains the methodology of the proposed GTESNet-S*. Section IV presents the experimental

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results of the tests conducted to evaluate the performance of GTESNet-S*. Finally, Section V concludes the article.

II. Related Work

A. Multimodal Scene Classification

Multimodal-based models aim to take full advantage of different data sources. He et al. [25] used two branches, local and global attention, to generate the correlation weights of the two input modes and then fuse the features adaptively. Liang et al. [26] introduced a fusion method that takes full advantage of each modality: using one branch corresponding to the features of both modalities, using one branch corresponding to the features of only one modality, and fusing the features of both branches. Tong et al. [27] presented a method for exploring and enhancing the modules with useful information at each stage of the fusion, which fully integrates the features. The aforementioned approaches attempt to fully utilize the information from multiple modalities to enhance the scene classification performance. Nevertheless, they cannot be straightforwardly applied to RSIs owing to the complex relationships between different modalities and large intra-class variation of objects.

Existing approaches have used various methods to address this problem. Zhu [28] designed a method for enhancing the multiscale features acquired by the transformer and extracting contextual features using a parallel structure built with convolutional layers. Zhu et al. [29] proposed a method that uses two parallel decoders to learn the change and semantic features separately and fuses the two using a fusion module to enhance the model performance. Wu et al. [30] proposed a method that combines edge features with prototypes to obtain better performance through an additional loss for enhanced supervision. Jiang et al. [31] used the ability of the attention mechanism to focus on target features effectively to enhance the performance of the U-Net model and improve its accuracy.

Although the abovementioned models for RSIs have achieved promising results, they may not be able to achieve a good trade-off between the performance and the number of parameters. Making models more lightweight has become a major concern, and the KD technique is considered promising owing to its strong potential for achieving model compression while maintaining performance.

B. Graph Reasoning Network

Graph convolutional networks (GCNs) can effectively model contextual information, which is convenient for scene classification. For example, Ma et al. [32] used a dual-component approach to characterize the correlation among pixels and the association within the channel dimension. Chai et al. [33] designed a graph-based structure to replace the transformer to capture long-range dependencies by exploiting the local relationships between pixels. Vekinis [34] proposed a structure that combines the architecture of a standard encoder–decoder and a two-branch GCN to augment features with semantic information at multiple spatial scales.

Inspired by the latter approach, we designed a new CGI model that directly mines contextual information within the target scope, and unlike in existing approaches, our operation is performed directly before the feature fusion.

C. KD

KD is a technique for transferring learnable knowledge from a complex teacher model to a simple student model, thereby significantly improving the performance of the student model. For example, Feng et al. [35] proposed a method for transferring knowledge from a multimodal model to an unimodal model that utilizes the multimodal boundary features of the teacher's network to facilitate student learning of the boundary information. Gao et al. [36] addressed the problem of imprecise pixel delineation of RSIs using the prediction frame of the teacher network to distinguish between different instances while removing instances that contain misleading knowledge. Xin et al. [37] proposed a means of selecting the k-nearest neighbors for each intermediate layer feature and final prediction, similar to the form of similarity maps to convey relational knowledge. Wu et al. [38] adopted a method for distilling predictions into relevant and irrelevant knowledge, and intermediate-level features into knowledge with high and low relevance.

In KD, incorrect knowledge from a teacher network can mislead the student network. Therefore, some approaches use multiteacher KD, which provides the student with additional guidance from multiple pretrained teachers with diverse sources of knowledge. Zhang et al. [39] used a meta-weighting module to assign proportions to diverse features from the output and intermediate layers of a teacher network to improve the performance of the students. Similarly, Chen et al. [40] adaptively assigned weights to the multiteacher model, thereby effectively enhancing the robustness of the distilled student model. Zhang et al. [41] assigned sample credibility to the prediction of each teacher with the help of true labels, whereby teachers with predictions close to the label are assigned larger credibility.

Inspired by the above methods, we attempt to use a simple confidence perception to assign different weights to the two teachers to exploit the advantages of dual-teacher KD. As opposed to previous studies, we use new CTD and GRD that enable GTESNet-S to learn the clustering, coding, and information mining abilities of the GTESNet-T. The classical and effective PDD is also employed to align the prediction of the GTESNet-S with that of the GTESNet-T.

III. Proposed Method

A. Overall Architecture

As shown in Fig. 2, the graph template enhancement and supplementation network (GTESNet) used an encoder–decoder structure. During encoding, ResNet34 [42] was used as the backbone for GTESNet-T to extract the RSI features because of its balanced extraction capability and the number of its own parameters, and MobileNet-v2 [43] was employed for GTESNet-S. The IRRG and nDSM images were input into two parallel encoders to obtain features R_i and D_i, where $i \in \{2, 3, 4, 5\}$ was the feature layer (because the features extracted in the first layer contained

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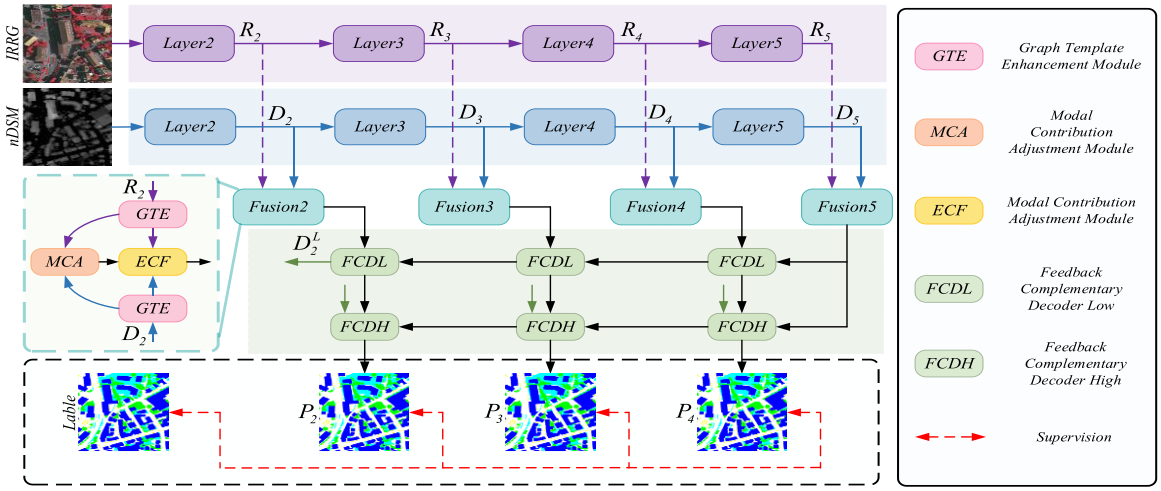


Fig. 2. Overall framework of the proposed GTESNet.

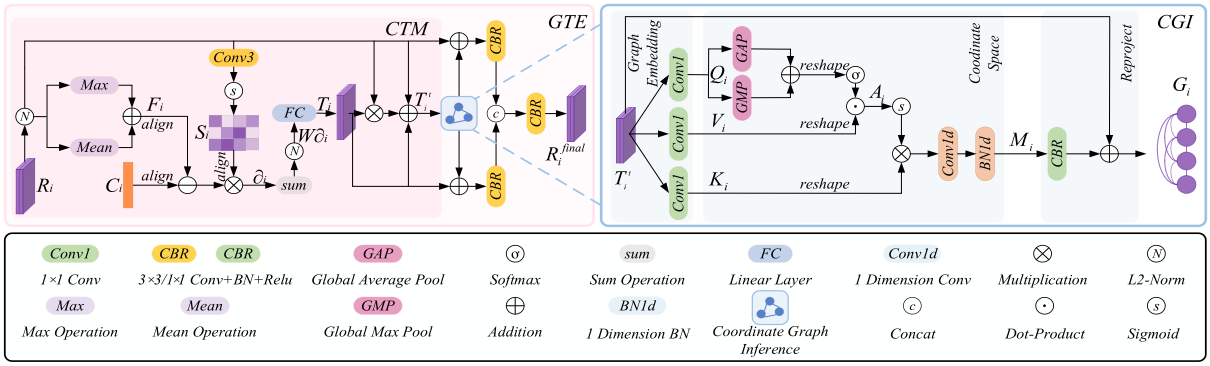


Fig. 3. Proposed GTE.

more interference and were not applicable in this study). Thereafter, the R_i and D_i were passed into the corresponding GTE, in which CTM was used to capture the common characteristics of objects in the same category to generate template features. Subsequently, CGI was employed to mine the long-range contextual information. The enhanced features were simultaneously fed into the MCA and ECF modules. The MCA assigned the proportions of features from different modalities to the next fusion process, thereby avoiding the problem of poor performance caused by an imbalance in the quality of images of different modalities. Subsequently, the proportions were sent to the ECF, where the features of both modalities were completely integrated. The fused features were sequentially sent to the first subdecoder FCDL in the designed FCD, and their outputs were passed to the second FCDH to complement the decoded features and generate the final prediction. Detailed information is presented in the below sections.

B. GTE Module

As shown in Fig. 3, the GTE included two main modules: CTM and CGI. The CTM focused on clustering the expressive features in the feature map to obtain templates that represented the features of the object. The templates comprised the corresponding template features, which were obtained from the clustering of the CTM and were used as inputs to the CGI to mine semantic information in the individual modal features.

For CTM, we created a learnable cluster-centered 2-D tensor, $C_i \in R^{k\times 1}$, as the centroids, where k was the number of templates in a template feature ($k = 24$ was the optimal value obtained experimentally). Thereafter, the input extracted features (e.g., R_i) were normalized. We used the $L2$-Norm because it improved the robust performance of the model. Next, we employed a 3×3 convolutional layer to ensure that the feature map had k channels to compute scores for each pixel in the mapping belonging to the k templates and employed the softmax operation to normalize the scores via

$$S_i = \mathrm{Conv}_{3\times 3}(\mathrm{Soft}(L2\mathrm{Norm}(R_i))) \tag{1}$$

where $\mathrm{Conv}_{3\times 3}$ is a convolutional operation with a 3×3 convolutional kernel, Soft is the softmax operation, and $L2$Norm is the $L2$-Norm. We transformed the scores into a tensor with the same spatial dimensions as the input features R_i and flattened it in HW dimensions via $S_i \in R^{k\times 1\times HW}$. We applied max and mean operations on the normalized features to obtain expressive feature mapping F_i by element-wise addition. We next extended the cluster-centered tensor C_i and the feature mapping to the same dimension to compute the residuals between each template and each pixel of the feature mapping F_i, hence, S_i' and $F_i' \in R^{k\times 1\times HW}$. The computed scores, S_i, were multiplied after the element-level subtraction operation, which is expressed as

$$F_i = \mathrm{Max}(L2\mathrm{Norm}(R_i)) + \mathrm{Mean}(L2\mathrm{Norm}(R_i)) \tag{2}$$

$$\partial_i = S_i \times (\mathrm{Align}(F_i) - \mathrm{Align}(Ci)) \tag{3}$$

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where Max and Mean are the maximum and mean operations, respectively; Align is the expand operation that equates the dimensions of C_i and F_i to meet the requirements of element-wise subtraction. The resulting residuals were summed along the HW dimension to obtain a weighted residual for each template, which had a shape of $R^{k\times 1}$. After applying the same $L2$-Norm for normalization, the weighted residuals were reshaped along the final dimensions as R^k using

$$W\partial_i = L2\text{Norm}(\text{Align}(\text{sum}(\partial_i)) \tag{4}$$

where sum is the sum operation. The weighted residuals, $W\partial_i$, reflect the magnitude of the change in the cluster-centered centroid, and it was continuously updated such that the change was 0 throughout training. Subsequently, the $W\partial_i$ values were transferred to a fully connected layer to reshape the output tensor into $R^{k\times c}$, which was expanded back into the same spatial dimension as the original input features to obtain T_i. Therefore, we obtained both the set of templates and the cluster-centered centroid

$$T_i = \text{Align}(\text{Linear}(W\partial_i)). \tag{5}$$

Furthermore, the template-enhanced feature, T_i^t, was obtained via the element-wise multiplication of the set T_i with the input feature R_i and their addition

$$T_i^t = T_i + R_i + (R_i \times T_i). \tag{6}$$

The output of the CTM was fed into the CGI, where we utilized the efficiency and effectiveness of graph convolution to exchange long-range information in a single layer. The graph convolution is defined as

$$G_i = \sigma(A_i X_i W_i) \tag{7}$$

where σ denotes the nonlinear activation function, A_i denotes the adjacency matrix representing the neighborhood relations of the graph, and W_i is the weight matrix. The input template-enhanced feature T_i^t was sent to three 1×1 convolutions as learnable linear transformations to project the input into a new coordinate space

$$V_i = \text{Conv}_1\left(T_i^t\right) \tag{8}$$

$$Q_i = \text{Conv}_1\left(T_i^t\right), \quad \text{and} \tag{9}$$

$$K_i = \text{Conv}_1\left(T_i^t\right). \tag{10}$$

The number of feature channels was halved using the above-mentioned operation to reduce the number of computations. Subsequently, we performed the global avepool and maxpool operations on Q_i and added them together. Through the reshape operation and the softmax function, the values were dot-multiplied with reshaped V_i to obtain an adjacency matrix A_i for diffusing information between nodes, which is given by

$$A_i = f(\text{Soft}\left((\text{gap}(Q_i) + \text{gmp}(Q_i))^{\text{Res}}\right), (V_i)^{\text{Res}}) \tag{11}$$

where gap and gmp are the global avepool and global maxpool operations, respectively, $()^{\text{Res}}$ is the reshape operation, and f is the dot product. Afterward, we multiplied the resulting value by reshaping K_i via a sigmoid operation. After 1-D convolution and batch normalization, we obtained the full-coordinate graph convolution M_i, which is expressed as

$$M_i = \text{BN1d}\left(\text{Conv1d}\left(\text{Soft}(A_i) \times (K_i)^{\text{Res}}\right)\right) \tag{12}$$

where Conv1d and BN1d are the 1-D convolution and 1-D batch-normalization, respectively. After reasoning, we projected the M_i back into the original space. We employed a simple 1×1 convolution followed by a rectified linear unit (ReLU) and batch normalization and residually connected it with the input features T_i^t to obtain the output map G_i via

$$G_i = \text{CBR}_{1\times 1}(M_i) + T_i^t \tag{13}$$

where $\text{CBR}_{1\times 1}$ is the 1×1 convolution followed by batch normalization and ReLU. Finally, we concatenated and convolved them to obtain the final output feature R_i^{final} to prevent interference from redundant information

$$R_i^{\text{final}} = \text{CBR}_{3\times 3}(\text{Cat}(\text{CBR}_{3\times 3}(G_i + R_i) \quad \text{CBR}_{3\times 3}(G_i + T_i))). \tag{14}$$

C. MCA Module

We proposed an MCA that adaptively adjusted the weights of the modalities and assigned contributions to modal fusion to address the issue of the uneven quality of two modal images for different scenes.

As shown at the bottom of Fig. 4, we used the outputs of the GTE as inputs in the MCA module. After using a 3×3 convolutional layer, we employed the global average pool operation on these to obtain average features Ave_i^R and $\text{Ave}_i^D \in R^{c\times 1\times 1}$ to ensure that the representation remains robust to disturbances. Subsequently, we fused the averaged features to generate the basis features, $B_i \in R^{c\times 1\times 1}$. These steps are expressed as

$$\text{Ave}_i^R = \text{gap}\left(\text{Conv}_{3\times 3}\left(R_i^{\text{final}}\right)\right) \tag{15}$$

$$\text{Ave}_i^D = \text{gap}\left(\text{Conv}_{3\times 3}\left(D_i^{\text{final}}\right)\right), \quad \text{and} \tag{16}$$

$$B_i = \text{MLP}\left(\text{MLP}\left(\text{Ave}_i^R \times \text{Ave}_i^D\right)\right) \tag{17}$$

where MLP is the multilayer perceptron. Thus, we obtained the contribution $(\beta_i, 1-\beta_i)$ that represented the participation of each modality in the fusion. Finally, we derived the cosine similarity β as

$$\beta_i = \frac{B_i^R}{B_i^R + B_i^D} \tag{18}$$

$$B_i^R = \text{Cosine}\left(B_i, \text{Ave}_i^R\right), \quad \text{and} \tag{19}$$

$$B_i^D = \text{Cosine}\left(B_i, \text{Ave}_i^D\right). \tag{20}$$

D. ECF Module

A major focus of research has been adequately fusing multimodal features to improve the model performance. As shown at the top of Fig. 4, the proposed ECF employed correlation matrices as the control mechanisms, and R_i^{final} and D_i^{final} denote the input features in layer i. We performed max and mean operations on R_i^{final} and D_i^{final} and summed the results to obtain f_i^R and $f_i^D \in R^{1\times H\times W}$ to reduce the number of computations. We next used three 1×1 convolutions and

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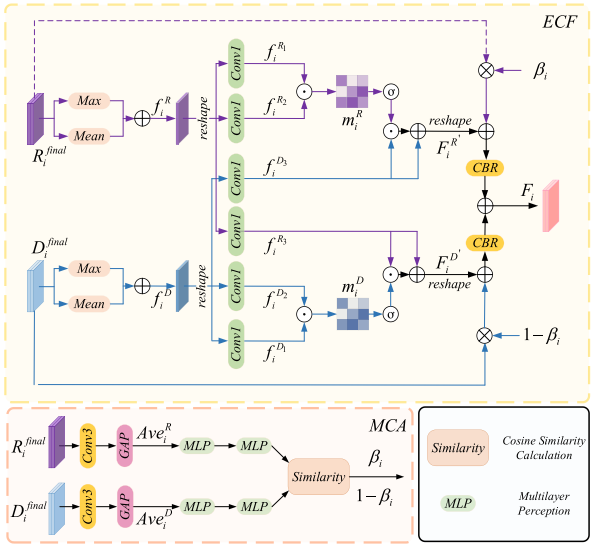


Fig. 4. Proposed ECF and MCA.

reshaped them to obtain different shapes for features $f_i^{R_j}$ and $f_i^{D_j}$, where $j \in \{1, 2, 3\}$. The equations used to perform these steps are

$$f_i^R = \max\left(R_i^{\text{final}}\right) + \text{mean}\left(R_i^{\text{final}}\right) \tag{21}$$

$$f_i^D = \max\left(D_i^{\text{final}}\right) + \text{mean}\left(D_i^{\text{final}}\right) \tag{22}$$

$$f_i^{R_j} = \left(\text{Conv}_{1\times1}\left(f_i^R\right)\right)^{\text{Res}}, \quad j = 1, 2, 3, \quad \text{and} \tag{23}$$

$$f_i^{D_j} = \left(\text{Conv}_{1\times1}\left(f_i^D\right)\right)^{\text{Res}}, \quad j = 1, 2, 3. \tag{24}$$

The elemental correlation matrix reflected the importance of the model at different positions. Therefore, we obtained the IRRG correlation matrix m_i^R and nDSM correlation matrix m_i^D via matrix multiplication of two features for the same modality using

$$m_i^R = f_i^{R_1} \cdot f_i^{R_2} \tag{25}$$

$$m_i^D = f_i^{D_1} \cdot f_i^{D_2} \tag{26}$$

where the dot represents matrix multiplication. The correlation matrix is widely used in the self-attention mechanism, and its memory occupation in previous models is considerably large (typically of size $m \in R^{HW\times HW}$), which is unsuitable for RSIs. In our ECF module, we effectively solved this problem by generating correlation matrices of size $m_i \in R^{H\times H}$.

To obtain complementary information from both modalities, we exchanged their correlation matrices

$$f_i^{R'} = \left(f_i^{D_3} \cdot \text{Soft}\left(m_i^R\right) + f_i^{D_3}\right)^{\text{Res}} \tag{27}$$

$$f_i^{D'} = \left(f_i^{R_3} \cdot \text{Soft}\left(m_i^D\right) + f_i^{R_3}\right)^{\text{Res}}. \tag{28}$$

Subsequently, we obtained enhanced features $f_i^{R'}$ and $f_i^{R'} \in R^{1\times H\times W}$. Next, we introduced the corresponding contributions β_i and $1-\beta_i$, which were generated by the MCA module to address the problem of uneven quality distribution and measure the relative importance of the two modalities. Specifically, we multiplied the raw input features by their corresponding contributions and performed a residual summation after generating the final fused features F_i. The

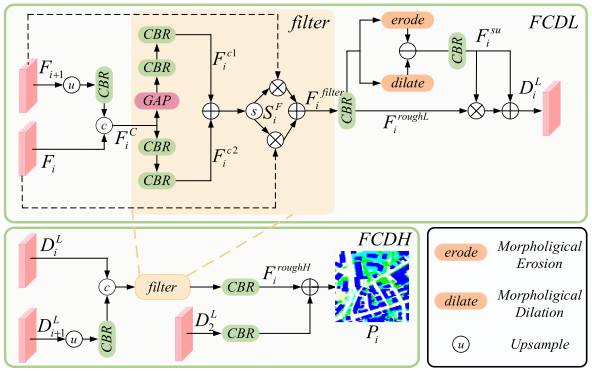


Fig. 5. Proposed FCDs.

representation is expressed as

$$F_i = \text{CBR}_{3\times3}\left(R_i^{\text{final}} \times \beta_i + f_i^{R'}\right) + \text{CBR}_{3\times3}\left(D_i^{\text{final}} \times (1-\beta_i) + f_i^{D'}\right). \tag{29}$$

E. FCDs

For multimodal scene classification, previous approaches have directly aggregated features at multiple levels and produced ultimate predictions. Distinctions among features exist at different levels; thus, direct aggregation may lead to missing or overlapping information, which may affect the final performance. Therefore, we designed FCDs consisting of two cascaded subdecoders that fed the features generated by the first decoder layer (FCDL) to the second layer (FCDH) to solve the problem of information loss, as shown in Fig. 5.

An adaptive filtering method was used to restrain superfluous information, avert contamination, and minimize the discrepancies among multiscale features. Compared to the direct aggregation of multilevel features used in previous studies, FCDL can eliminate background disruptions and sharpen feature boundaries. First, the module performed upsampling and a 3×3 convolution to equate the size of the high-level feature F_{i+1} to that of the low-level feature F_i. Next, we fed F_i^C into the filter we designed, in which the feature was subjected to only two consecutive 1×1 convolution blocks to obtain F_i^{C2} in one branch. In the other branch, F_i^C was first subjected to a global average pooling operation, and subsequently, it went through the same convolution blocks to obtain F_i^{C1}, which halved the number of channels. The results were added together, and the attention graph S_i^F was generated after a sigmoid operation. The input features were filtered by multiplying them with the attention map to get F_i^{filter}. The entire process is given by

$$F_i^C = \text{Cat}(\text{CBR}_{3\times3}(\text{Up}(F_{i+1})), F_i) \tag{30}$$

$$F_i^{C1} = \text{CBR}_{1\times1}\left(\text{CBR}_{1\times1}\left(F_i^C\right)\right) \tag{31}$$

$$F_i^{C2} = \text{CBR}_{1\times1}\left(\text{CBR}_{1\times1}\left(\text{gap}\left(F_i^C\right)\right)\right) \tag{32}$$

$$S_i^F = \text{sigmoid}\left(F_i^{C1} + F_i^{C2}\right) \tag{33}$$

$$F_i^{\text{filter}} = S_i^F \times F_{i+1} + \left(1 - S_i^F\right) \times F_i \tag{34}$$

where Up is the upsampling operation. We further extracted the F_i^{filter} using a 3×3 convolutional block to obtain a rough prediction feature F_i^{roughL}. Next, we used the rough prediction

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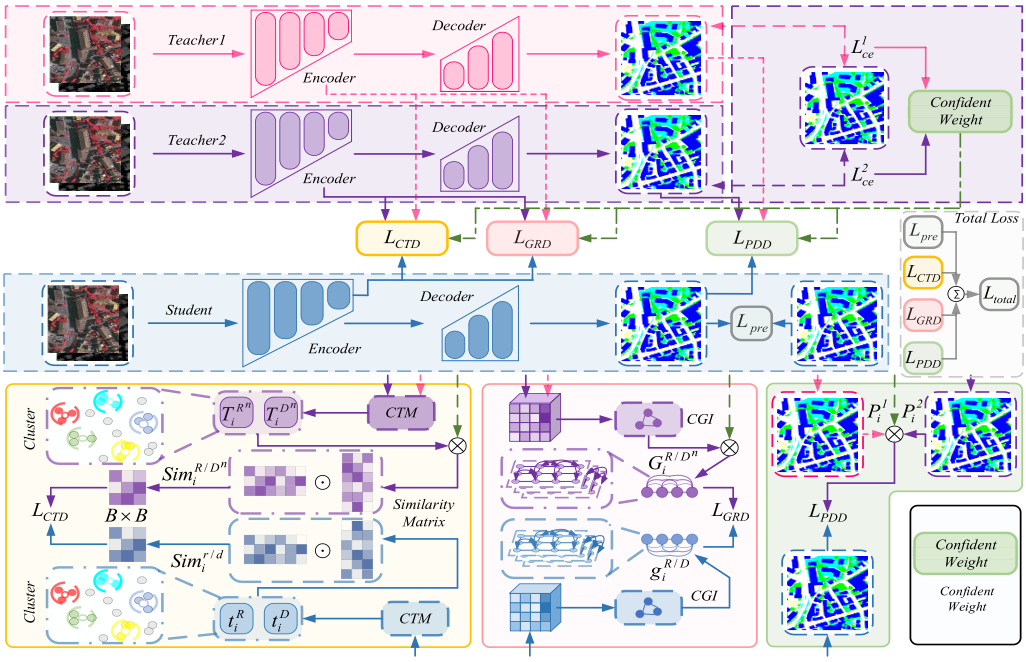


Fig. 6. Proposed KD strategy, which includes three parts: CTD, GRD, and FDD.

result to obtain the supplement features F_i^{su} via dilation and erosion operations, which are expressed as

$$F_i^{\mathrm{roughL}} = \mathrm{CBR}_{3\times3}\left(F_i^{\mathrm{filter}}\right) \tag{35}$$

$$F_i^{\mathrm{su}} = \mathrm{CBR}_{3\times3}\left(\mathrm{dil}\left(F_i^{\mathrm{roughL}}\right) - \mathrm{ero}\left(F_i^{\mathrm{roughL}}\right)\right) \tag{36}$$

where Up is the upsampling operation, and dil and ero are two opposite morphological operations: dilation and erosion. These two operations began by padding the perimeter of the input features with zeros to prevent their size from recessing after the operation. The second and third dimensions of the features were sequentially expanded using an unfold operation to convert the feature into a patch. The two operations were only performed oppositely in the last step: the dilation operation took the maximum value of the last dimension of the patch, whereas the erosion operation took the minimum value of the last dimension of the patch. This procedure is represented as

$$P = \mathrm{Unf}\left(\mathrm{Unf}\left(\mathrm{Pad}\left(F_i^{\mathrm{roughL}}\right)\right)\right)$$
$$P_m = (P)^{\mathrm{Res}}, \quad P_{\mathrm{di}} = \max(P_m), \quad P_{\mathrm{er}} = \min(P_m) \tag{37}$$

where the shape of P is $R^{B\times C\times H\times W\times3\times3}$, the shape of P_m is $R^{B\times C\times H\times W\times9}$, P_{di}, $P_{\mathrm{er}} \in R^{B\times C\times H\times W}$, Pad is the pad operation, and Unf is the unfold operation. We multiplied the supplement feature F_i^{su} with the input F_i^{roughL} and performed residual connections to add detail to the feature

$$D_i^L = F_i^{\mathrm{su}} \times F_i^{\mathrm{roughL}} + F_i^{\mathrm{su}}. \tag{38}$$

In the first decoder layer, the fused and decoded features of the FCDL were sent from deep to shallow layers to the upper-level FCDL, and their outputs were sent as inputs to the FCDH of the second decoder layer. In the FCDH, the structure preceding the filter was the same as that following the filter. The difference was that the FCDH directly received the output D_2^L from the shallowest layer of the DCDL as a compliment

$$D_i^H = F_i^{\mathrm{roughH}} + \mathrm{CBR}\left(D_2^L\right). \tag{39}$$

We utilized labels to monitor the outputs of all FCDHs to enhance the model performance.

F. GTESNet-S*

The overall structure of the dual-teacher KD approach we designed is illustrated in Fig. 6. A KD strategy integrating the knowledge of the two teachers was designed to adaptively involve them in the training process of guiding the student networks and improve the resilience and accuracy of the trained student networks and enhance their ability to cope with misleading knowledge from teachers. We adopted the work of Chen et al. [40] and Zhang et al. [41] and designed a simple adaptive weight assignment algorithm based on teacher performance to assign reasonable weights to the two teachers.

We designed GTESNet-S* (GTESNet-S with KD), which consisted of three distillation components: CTD, GRD, and PPD. The primary function of CTD enabled GTESNet-S to mimic the clustering ability of the two GTESNet-Ts. GRD was designed to stimulate the ability of GTESNet-S to mine long-range contextual information. PPD allowed GTESNet-S to continuously mimic the prediction results of GTESNet-T and improve its prediction performance.

1) Confidence Weight: We utilized the calculated cross-entropy loss between the labels and the ultimate predictions of the two teachers to assign higher weights to the better-performing teacher to capitalize on the respective strengths of the two teachers, which is expressed as

$$L_{\mathrm{CE}} = -\sum_{c=1}^{C} Y^c \times \log\left(P^c\right) \tag{40}$$

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$$L_{\mathrm{CE}}^{n} = L_{\mathrm{CE}}\left(P_2^n, \mathrm{label}\right) \tag{41}$$

$$w^n = \frac{1}{N-1}\left(1 - \frac{e^{L_{\mathrm{CE}}^n}}{\sum_{n=1}^{N} e^{L_{\mathrm{CE}}^n}}\right) \tag{42}$$

where L_{CE} is the cross-entropy loss function, C is the number of categories, Y^c is an indicator variable that equals 1 if the predicted class matches the sample class, and 0 otherwise, and N is the number of teachers ($N = 2$ in our model).

2) CTD: Most KD methods start with the following. Transferring knowledge of features at the same level or at different levels of the same modality prompts students to learn the teacher's powerful ability to mine information. Compared to previous studies, we started from a new perspective: learning the clustering process of GTESNet-T. Specifically, the extracted features R_i^n and D_i^n in GTESNet-T can be obtained as template features $T_i^{R^n}$ and $T_i^{D^n}$ that contain rich clustering information after passing through the CTM module in the GTE module proposed in this study. Similarly, t_i^R and t_i^D were obtained using GTESNet-S using the above operation.

After obtaining two sets of teacher template characteristics and one set of student template characteristics, we constructed the pairwise activation similarity matrices [44] Sim_i^{R/D^n} and $\mathrm{Sim}_i^{r/d} \in R^{B\times B}$ for T_i^{R/D^n} and $t_i^{R/D}$, which are given by

$$\mathrm{Sim}_i^{R/D^n} = T_i^{R/D^n} \cdot \left(T_i^{R/D^n}\right)^{\mathrm{T}} \tag{43}$$

$$\mathrm{Sim}_i^{r/d} = t_i^{R/D} \cdot \left(t_i^{R/D}\right)^{\mathrm{T}} \tag{44}$$

where T_i^{R/D^n} and $t_i^{R/D}$ are the template features of the nth teacher and student, respectively, which were uniformly sized by the convolutional block. Constructing inputs in this manner allowed each teacher and student input to have the same dimensions. The $b \times b$ matrix encoded the semantic similarity information for each teacher–student sample pair and did not vary with the rotation of the feature space. Subsequently, we used the mean square error (mse) loss to compute the distance among clustered competencies while using the resulting confidence to assign weights to the teachers

$$L_{\mathrm{CTD}} = \sum_{n=1}^{N} w^n L_{\mathrm{MSE}}\left(\mathrm{Sim}_i^{R/D^n}, \mathrm{Sim}_i^{r/d}\right) \tag{45}$$

$$L_{\mathrm{MSE}}(x, y) = \frac{\sum_{h,w=1}^{HW}(x_{hw}, y_{hw})^2}{HW} \tag{46}$$

where HW is the total number of pixels in the feature map's spatial dimension.

3) GRD: In our model, the template features were directly sent to the CGI module, which was designed to use graph reasoning networks (GRNs) [32] to visually model pixel-level relationships and capture global contextual information. Thus, we used the potential of the student network to mine richer semantic information for the final prediction. Specifically, we designed a dice-based relational KD in each layer for the features G_i^{R/D^n} and $g_i^{R/D}$, which were processed by the CGI module to enhance the semantic features through

$$L_{\mathrm{GRD}} = \sum_{n=1}^{N} w^n L_{\mathrm{dice}}\left(G_i^{R/D^n}, g_i^{R/D}\right) \tag{47}$$

$$L_{\mathrm{dice}}(x, y) = 1 - \frac{2\sum_{i=1}^{\mathrm{total}} s_i t_i}{\sum_{i=1}^{\mathrm{total}} s_i + \sum_{i=1}^{\mathrm{total}} t_i} \tag{48}$$

where total is the sum of the number of pixels in the image (total $= H \times W \times B$).

4) PDD: In FCDs, the prediction results of the first subdecoder were used as supervisory inputs to the second subdecoder to complement and correct its outputs and improve the performance of the model. Thus, the second layer of decoders reflected the predictive power of our model. Therefore, we used a simple PDD that used the Kullback–Leibler divergence (KLD) to mimic the predictive power of GTESNet-T, which is expressed as

$$L_{\mathrm{PDD}} = \sum_{n=1}^{N} w^n L_{\mathrm{KLD}}\left(P_i^n, p_i\right). \tag{49}$$

G. Total Loss

Labels were used to progressively supervise the three-layer predictions of the FCDHs, which were denoted as p_i ($i = 2$, 3, 4). Cross-entropy was used to compute the prediction loss for pixel-level surveillance, which is given by

$$L_{\mathrm{pre}} = \sum_{i=1}^{3} L_{\mathrm{CE}}(p_i, \mathrm{label}). \tag{50}$$

After training the two GTESNet-Ts, their weights were fixed and the student network, GTESNet-S, was refined. The total loss of GTESNet-S* is expressed as

$$L_{\mathrm{total}} = L_{\mathrm{pre}} + L_{\mathrm{CTD}} + L_{\mathrm{GRD}} + L_{\mathrm{PDD}}. \tag{51}$$

IV. Experiments

A. Datasets and Evaluation Metrics

We evaluated GTESNet-S* on two publicly available datasets: the Vaihingen and Potsdam datasets. Both datasets cover six categories: impervious surfaces, buildings, low vegetation, trees, cars, and clusters/backgrounds. In addition to 16 highly labeled images, IRRG images consisting of near-infrared, green, and red with dimensions of 2500 × 2500 pixels, as well as the associated nDSMs, are included in the Vaihingen dataset. We divided the dataset into two sections: 11 images for training and five images for validation. In contrast to the former, apart from 24 highly labeled images of 6000 × 6000 pixels, the Potsdam dataset also includes IRRG and nDSM images. From these, we selected seven images as the validation set and the remainder were used to train the model. Owing to the large image sizes in both datasets, they were cropped to 256 × 256 pixels.

We mainly adopted the accuracy (Acc), intersection over union (IoU), mean accuracy (mAcc), and mean IoU (mIoU) to evaluate the model performance in scene classification [45], [46]. The assessment was performed with uniformly sized 256 × 256 pixel images.

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TABLE I
PERFORMANCE COMPARISON BASED ON VAIHINGEN DATASET, WHERE THE OPTIMAL VALUES IN EACH COLUMN ARE COLORED IN RED AND THE SUBOPTIMAL VALUES IN GREEN

	Imp.surf		Building		Low veg.		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$[8]	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
FCN-8S$_{2015}$[5]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
SegNet$_{2017}$[4]	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
DeepLabv3+$_{2018}$[6]	90.06	81.11	87.04	82.70	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
RedNet$_{2018}$[17]	91.49	84.62	94.81	91.07	78.67	66.59	91.41	78.27	59.77	56.06	83.23	75.32
ACNet$_{2019}$[10]	91.95	85.34	95.45	91.82	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
HRCNet$_{2021}$[12]	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
ESANet$_{2021}$[9]	92.09	85.18	94.93	91.16	75.72	65.48	92.35	77.65	75.92	70.11	86.20	77.92
UNetFormer$_{2022}$[18]	91.74	85.26	95.36	91.89	78.69	67.33	91.89	78.63	82.98	79.30	88.13	80.48
FTUNetFormer$_{2022}$[18]	89.42	78.02	89.34	83.88	70.78	60.15	91.12	74.09	53.42	50.73	78.82	69.38
CMXNet$_{2022}$[11]	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
MANet$_{2022}$[7]	91.57	83.23	93.77	90.04	76.29	65.59	91.89	77.94	86.03	81.01	87.91	79.56
CMANet$_{2022}$[16]	92.02	85.06	95.29	91.15	77.35	66.06	91.48	78.54	73.56	68.89	85.94	77.94
ABCNet$_{2022}$[23]	86.60	80.49	93.55	87.87	80.77	63.72	87.24	74.16	18.14	17.90	73.26	64.83
MGCNet$_{2022}$[19]	91.18	84.89	95.58	91.52	76.22	65.74	91.76	78.53	79.56	75.33	87.06	79.20
A2FPN$_{2022}$[20]	92.12	82.54	93.64	90.59	77.36	65.58	90.36	78.82	89.82	81.04	88.80	79.61
SFAF-MA$_{2023}$[25]	90.94	85.37	95.67	92.21	80.79	67.55	90.78	78.95	74.74	71.19	86.62	79.05
CATNet$_{2023}$[21]	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
EAEFNet$_{2023}$[26]	93.78	84.37	95.35	92.25	79.87	67.98	89.11	79.00	81.54	74.24	87.93	79.57
CEKD$_{2023}$[35]	92.22	85.05	95.78	91.66	76.57	66.77	92.69	78.97	76.63	72.01	86.71	78.89
CPNet$_{2024}$[22]	91.79	78.58	93.99	88.97	61.96	55.58	90.77	74.31	29.63	27.68	74.23	65.02
GTESNet-T1	92.49	86.28	96.37	91.88	79.08	68.20	91.90	79.44	88.40	82.36	89.65	81.83
GTESNet-T2	91.32	85.69	95.76	92.85	80.81	68.07	91.10	78.91	88.81	83.00	89.56	81.70
GTESNet-S	91.10	83.88	94.72	91.95	73.93	65.18	90.57	78.76	75.08	68.80	86.08	77.71
GTESNet-S*	92.36	85.89	95.61	92.24	78.19	68.23	92.73	79.52	86.87	80.78	89.15	81.33

B. Implementation Details

All experiments were conducted in a PyTorch environment using an NVIDIA TITAN V graphics card and 16 GB of RAM. Before starting the training, the input images were flipped and randomly cropped to augment the training samples. The training was conducted in two phases. In the first stage, after the backbone was initialized according to ResNet-34, we trained GTESNet-T for 120 epochs under label-only supervision. This training was performed twice to obtain the weights of the two groups of teachers. In the second stage, using the same training method, after initializing the weights in MobileNet-V2, we fixed the weights of the two GTESNet-Ts and trained GTESNet-S for 120 epochs using our proposed L_{total} as supervision.

The Adam optimizer was used and the initial learning rate was set to 0.0001, accompanied by a weight decay of 0.0005. The network batch sizes were 16 for the student training and 10 for the teacher training.

C. Comparisons With SOTA Methods

We compared it with the following 21 SOTA methods to evaluate the performance of GTESNet-S*: U-Net [8], FCN-8S [5], SegNet [4], DeepLabv3+ [6], RedNet [17], ACNet [10], HRCNetx [12], ESANet [9], UNetFormer [18], FTUNetFormer [18], CMXNet [11], MANet [7], CMANet [16], ABCNet [23], MGCNet [19], A2FPN [20], SFAF-MA [25], CATNet [21], EAEFNet [26], CEKD [35], and CPNet [22].

Tables I and II list the quantitative results for the two datasets, including the general performance and performance for each category. In summary, the proposed GTESNet-S* achieved excellent outcomes based on all evaluation metrics. The two GTESNet-Ts outperformed the comparison methods in most metrics on both datasets. On the Vaihingen dataset, GTESNet-T achieved eight firsts and three seconds. On the Potsdam dataset, it achieved seven firsts and seven seconds, and its performances in terms of the remaining metrics were comparable to the best performances for each metric. The proposed GTESNet-T performed better on the Vaihingen dataset than on the Potsdam dataset. Our model distinguished the categories of impervious surfaces, buildings, cars, and clusters more accurately, and the results in these categories were significantly better than those of the other methods. This demonstrates that our model can process large and small objects because it clusters similar features of objects in the same category to increase their differentiation, and the mining of semantic information by CGI enhances the results. The overall performances of the two teachers were identical; however, significant differences were observed in each category. The KD method effectively solved this problem by delivering relatively high-quality knowledge to students, enabling the originally poorly performing student network to achieve extraordinary performance while retaining its lightweight. However, RSIs usually involve different complex scenes, and in a few cases, high objects, such as buildings, and trees, can obscure low vegetation and impervious surfaces located in low areas, leading to problems with pixel segmentation in this part of the region, and thus the classification performance of certain objects is slightly lower than that of SOTA methods.

Fig. 7 presents the scene classification results for a selection of the models, demonstrating the performance of GTESNet-S*. Rows 1–4 show the classification results for the four groups of models selected from the Vaihingen dataset, whereas rows 5–7 show the classification results for the three groups of models selected from the Potsdam dataset. The images indicate that GTESNet-S* can accurately and efficiently classify scenes for different scenarios.

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TABLE II
PERFORMANCE COMPARISON BASED ON POTSDAM DATASET

	Imp.surf		Building		Low veg.		Tree		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
U-Net$_{2015}$[8]	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	93.89	90.24	50.30	36.26	82.13	72.22
FCN-8S$_{2015}$[5]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
SegNet$_{2017}$[4]	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
DeepLabv3+$_{2018}$[6]	91.57	82.49	91.78	87.59	87.36	73.63	85.45	73.32	93.89	90.04	53.80	43.54	83.97	75.10
RedNet$_{2018}$[17]	92.19	82.83	93.61	90.13	87.00	73.22	83.00	71.77	93.36	90.08	56.74	43.51	84.32	75.26
ACNet$_{2019}$[10]	91.32	82.74	93.83	90.06	86.16	73.53	86.03	72.87	93.79	90.43	54.51	41.65	84.27	75.21
HRCNet$_{2021}$[12]	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
ESANet$_{2021}$[9]	91.38	82.92	93.69	89.82	87.10	73.16	82.48	70.81	93.08	88.53	55.68	43.38	83.90	74.77
UNetFormer$_{2022}$[18]	91.07	81.76	91.87	87.05	84.84	72.87	86.83	73.18	94.81	89.42	55.27	41.05	84.11	74.22
FTUNetFormer$_{2022}$[18]	87.77	77.93	87.98	81.38	86.68	68.23	72.06	59.72	92.40	87.52	44.98	35.54	78.65	68.39
CMXNet$_{2022}$[11]	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
MANet$_{2022}$[7]	90.48	81.20	87.58	83.47	88.78	72.74	82.69	71.54	90.96	88.73	46.16	38.11	81.11	72.63
CMANet$_{2022}$[16]	91.75	81.87	93.64	90.23	86.42	73.59	84.70	72.40	92.74	89.11	55.97	42.74	84.20	74.99
ABCNet$_{2022}$[23]	91.77	77.86	89.20	81.75	87.62	72.02	78.31	70.09	91.91	87.74	31.89	29.70	78.45	69.86
MGCNet$_{2022}$[19]	90.61	81.21	93.87	89.02	86.72	72.67	83.70	72.51	93.39	88.91	53.24	39.66	83.59	74.00
A2FPN$_{2022}$[20]	90.05	82.49	90.82	85.52	86.69	73.29	85.99	73.36	94.54	90.26	55.28	41.72	83.89	74.49
SFAF-MA$_{2023}$[25]	90.63	82.06	93.31	89.67	85.76	72.35	83.51	70.95	92.55	88.34	59.03	42.40	84.30	74.30
CATNet$_{2023}$[21]	88.12	79.80	93.32	89.52	85.91	70.66	78.78	65.85	91.56	84.99	55.58	40.41	82.21	71.87
EAEFNet$_{2023}$[26]	92.32	82.04	93.22	89.54	84.34	72.37	85.02	71.91	94.64	88.30	53.55	39.34	83.85	73.92
CEKD$_{2023}$[35]	91.09	80.89	93.37	88.97	84.70	71.71	85.18	71.62	93.01	89.35	52.63	38.18	83.33	73.45
CPNet$_{2024}$[22]	89.36	75.08	87.59	85.02	85.99	67.01	73.81	60.45	39.93	38.42	36.54	25.91	68.87	58.65
GTESNet-T1	91.77	83.07	93.47	90.09	86.15	73.71	85.46	73.96	94.73	89.19	59.84	43.58	85.24	75.60
GTESNet-T2	90.16	83.22	93.77	90.30	88.10	74.12	84.32	73.10	95.02	89.32	59.60	45.03	85.16	75.85
GTESNet-S	88.85	80.53	92.86	89.10	85.40	71.34	83.95	70.78	91.77	87.70	54.02	39.68	82.98	73.19
GTESNet-S*	91.13	82.75	93.56	90.21	85.07	73.15	86.32	74.02	93.78	89.71	60.88	42.02	85.12	75.31

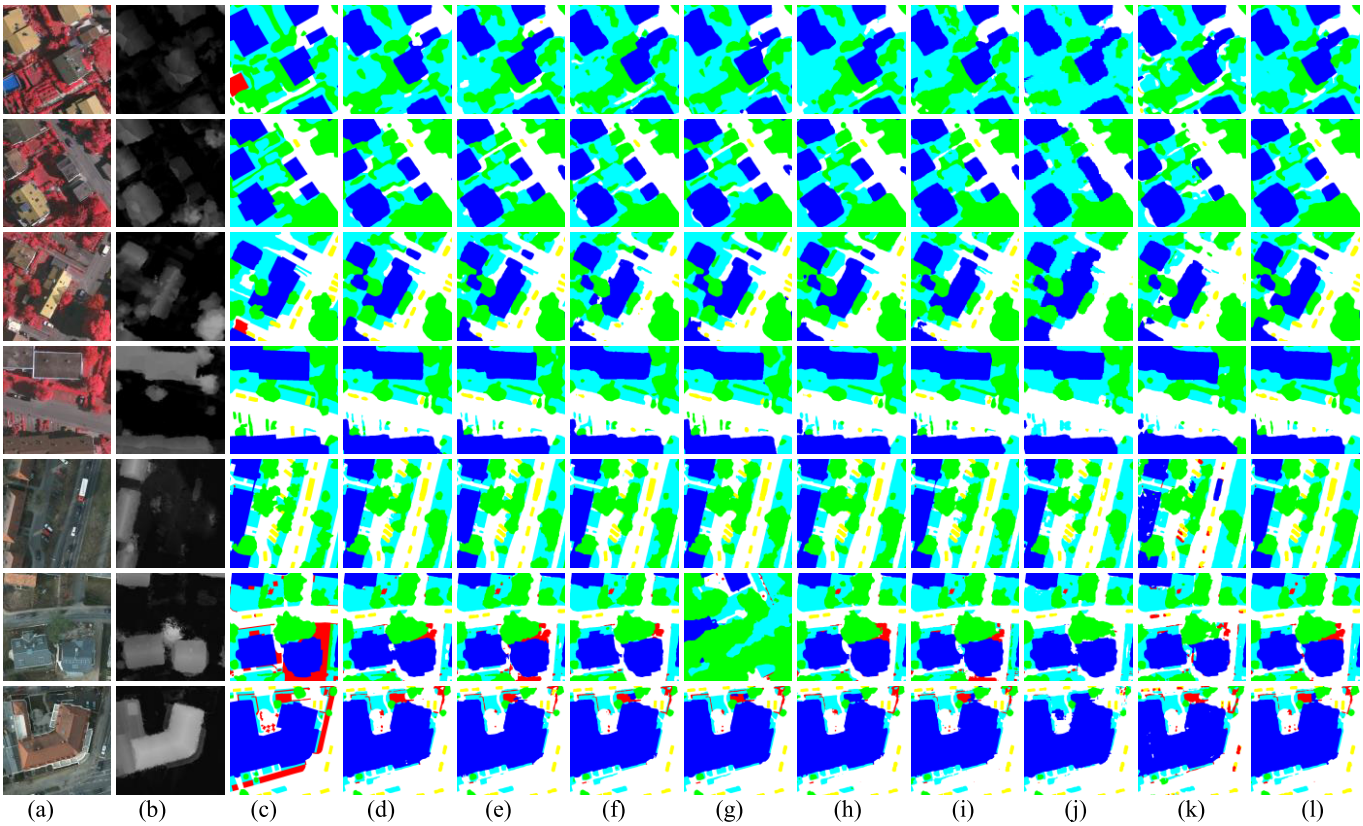


Fig. 7. Prediction results: (a) IRRG, (b) nDSM, (c) label, (d) GTESNe-T1, (e) GTESNe-T2, (f) REDNet, (g) ACNet, (h) ESANet, (i) CMXNet, (j) ABCNet, (k) CPNet, and (l) GTESNet-T*.

Table III compares the complexity and training speed of each model according to three indicators: the numbers of parameters, floating-point operations (FLOPs), and the average time to train one epoch on the Vaihingen dataset with a batch size of 10. GTESNet-S* has a parameter size of 16.02 M, which is considerably smaller than that of the other models. In addition, our model required only 3.13 Gmac of FLOPs, proving the computational efficiency of GTESNet-S*. Moreover, the student network was very fast when trained alone. After adding the teacher network as supervision again, the training speed of the model decreased slightly but stayed good.

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TABLE III
COMPARISON OF BACKBONE, NUMBER OF PARAMETERS, AND RUNNING SPEED FOR DIFFERENT MODELS

Model	Encoder	Params (M)	Flops (Gmac)	Time (S)
U-Net$_{2015}$[8]	VGG-16	26.36	55.93	399
FCN-8S$_{2015}$[5]	VGG-16	134.29	74.55	891
SegNet$_{2017}$[4]	VGG-16	53.56	18.3	315
DeepLabv3+$_{2018}$[6]	ResNet-101	59.33	32.45	579
RedNet$_{2018}$[17]	ResNet-50	81.95	21.17	351
ACNet$_{2019}$[10]	ResNet-50	116.6	26.41	438
HRCNet$_{2021}$[12]	ResNet-50	62.71	30.27	407
ESANet$_{2021}$[9]	ResNet-34	45.42	10.15	285
FTUNetFormer$_{2022}$[18]	Swin-Transformer	36.17	16.85	311
CMXNet$_{2022}$[11]	SegFormer	66.56	14.26	332
MANet$_{2022}$[7]	ResNet-50	35.86	19.49	297
CMANet$_{2022}$[16]	ResNet-50	117.82	28.61	577
MGCNet$_{2022}$[19]	ConvNeXt	209.99	47.51	878
SFAF-MA$_{2023}$[25]	ResNet-50	232.08	18.24	742
CATNet$_{2023}$[21]	Swin-Transformer	262.74	76.79	1178
EAEFNet$_{2023}$[26]	ResNet-50	77.97	23.3	354
CEKD$_{2023}$[35]	ResNet-50	245.03	68.61	776
CPNet$_{2024}$[22]	Swin-Transformer	222.91	57.56	916
GTESNet-T	ResNet-34	76.33	17.95	315
GTESNet-S	MobileNet-v2	16.02	3.13	270
GTESNet-S*	MobileNet-v2	16.02	3.13	412

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON KD STRATEGY

Models	CTD	GRD	PDD	V		P	
				mAcc	mIoU	mAcc	mIoU
T1				89.65	81.83	85.24	75.60
T2				89.56	81.70	85.16	75.85
S				86.08	77.71	82.98	73.19
T1	√			86.47	78.59	83.28	73.87
T2	√			86.79	78.51	83.73	73.54
T1+T2	√			86.92	78.68	84.26	73.91
T1		√		86.34	78.41	83.65	73.59
T2		√		86.63	78.53	83.56	73.53
T1+T2		√		86.81	78.93	84.21	73.72
T1			√	87.60	79.50	83.60	73.84
T2			√	87.72	79.87	83.75	73.95
T1+T2			√	88.09	80.21	84.09	74.38
T1	√	√		87.10	79.06	84.45	74.29
T2	√	√		87.16	79.17	84.27	74.34
T1+T2	√	√		87.45	79.73	84.57	74.80
T1	√	√	√	88.30	80.84	84.77	75.19
T2	√	√	√	88.27	80.81	84.49	75.22
T1+T2	√	√	√	89.15	81.33	85.12	75.31

D. Ablation Study

We conducted ablation studies on the main modules and KD strategy under the same training settings to prove the validity of each component in GTESNet-S*. Table IV outlines the performance of our GTESNet-S* on the two datasets supervised by different teachers with different KD losses, where the √ symbol indicates that the corresponding KD losses were used. T1 indicates that only GTESNet-T1 supervised the student and T1 + T2 indicates that both teachers supervised simultaneously. Table V shows the impact of each important module in GTESNet-S* on the prediction results in the presence of KD policy supervision. We employed the w/o approach, removing one module at a time. As both CTD and GRD in this KD strategy were intricately linked to the GTE module, the effectiveness of GTE was explored without KD strategy supervision. Table VI demonstrates how a hyperparameter k in the learnable tensor created by the CTM affects the model performance, where the k value indicates that k templates exist in a template feature.

1) Validity of KD Strategy: We conducted a stepwise cumulative ablation experiment with three distillation losses and explored how having either one or two teachers affected the results to investigate the plausibility of the KD strategy, as shown in Table IV. The performance of the student network was gradually enhanced by the progressive accumulation of KD losses. Specifically, on the Vaihingen dataset, the total KD strategy increased the performance of the student network by 3.07% and 3.62% in the mAcc and mIoU metrics, respectively, with the performance of GTESNet-S, thus approaching that of the teacher. Compared with one teacher, when acting together, the dual teachers enhanced the model performance most obviously when CTD, GRD, and PDD were involved. This trend could be observed in the Potsdam dataset, where the two metrics of the student networks increased by 2.14% and 2.21%, respectively. In addition, the effect of dual teachers was significantly higher than that of the single teacher, owing to our KD strategy. Fig. 8 shows that the KD strategy improves the scene classification accuracy.

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON MODULES

Models	V		P	
	mAcc	mIoU	mAcc	mIoU
T1	89.65	81.83	85.24	75.60
T2	89.56	81.70	85.16	75.85
S	86.08	77.71	82.98	73.19
S*	89.15	81.33	85.12	75.31
S w/o GTE	83.97	75.77	82.47	72.10
S* w/o MCA	88.68	80.95	84.65	75.10
S* w/o MCA+ECF	86.86	79.85	84.07	74.41
S* w/o FCDL	87.47	79.66	83.36	74.15
S* w/o FCDH	88.20	80.37	84.45	75.20
S* w/o FCDL+FCDH	86.93	79.38	83.29	74.00

TABLE VI
RESULTS OF ABLATION EXPERIMENTS ON GTE

CLUSTER k	V		P	
	mAcc	mIoU	mAcc	mIoU
6	86.70	78.06	84.49	74.16
12	88.00	80.45	84.28	74.34
24	89.15	81.33	85.12	75.31
36	87.05	79.58	84.03	74.90

2) Validity of GTE: According to Table V, our proposed GTE significantly affected the final prediction of the model. Removing the GTE decreased the mAcc and mIoU by 2.11% and 1.94%, respectively, on the Vaihingen dataset and by 1.44% and 1.75%, respectively, on the Potsdam dataset. Fig. 9 shows the impact of GTE on the predictions. After removing the GTE, the outlines of certain objects in the image that were in contact with one another changed, and the pixel classifications in these locations were inaccurate. Table VI shows that the learnable clustering center tensor in the GTE significantly affected the final performance of the model, with the k value determining the number of templates in

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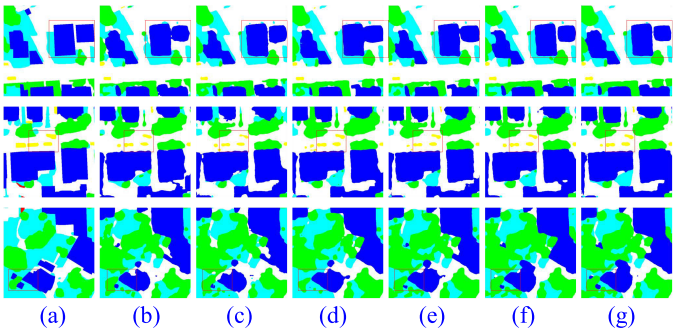


Fig. 8. Scene classification outcomes of validity of KD methods: (a) IRRG, (b) with CTD, (c) with GRD, (d) with PDD, (e) with CTD and GRD, (f) single teacher GTESNet-S*, and (g) GTESNet-S*.

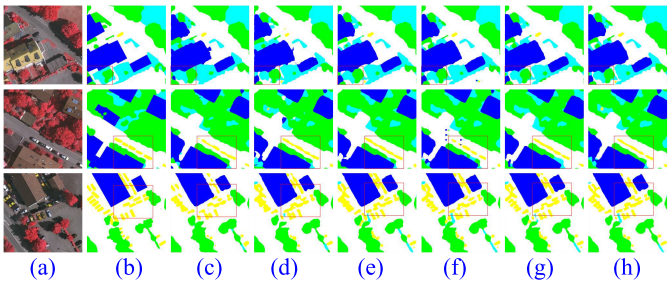


Fig. 9. Scene classification outcomes of validity of GTE: (a) IRRG, (b) label, (c) GTESNet-S without GTE, (d) GTESNet-S, (e) GTESNet-S*, (f) GTESNet-S* ($k = 6$), (g) GTESNet-S* ($k = 12$), and (h) GTESNet-S* ($k = 36$).

a template feature. We conducted ablation experiments to determine the optimal value of k to be 24, whose metrics were 1.15% and 0.88% higher than the next best on the Vaihingen dataset and 0.63% and 0.41% higher on the Potsdam dataset. In addition, this confirmed that our GTE can indeed increase the distinguishability of different categories of features while mining contextual information to improve the aggregate classification ability most effectively. Fig. 9 shows that removing the GTE module degraded the model performance and that $k = 24$ was the best choice for achieving optimal scene classification results.

3) Effectiveness of MCA and ECF: As shown in Table V, after directly removing the MCA, which is a module that assigns weights to modes, both the mAcc and mIoU metrics moderately decreased. After removing the MCA and replacing the ECF module with the simplest element-level think-aloud, our module produced a significant drop: 2.29% and 1.48% for the metrics on the Vaihingen dataset, and 1.05% and 0.90% for both on the Potsdam dataset, which demonstrated that our ECF effectively interacted with the information of the two modalities to realize a fusion module that was useful for scene categorization. This was also evidenced by the drop in the effect of the predicted images in Fig. 10 without the MCA and ECF.

4) Validity of FCDs: As shown in Table V, with the cascade structure preserved, FDCL boosted the results slightly more than FCDH: 0.73% and 0.71% higher on the Vaihingen dataset, whereas the values on the Potsdam dataset screwed up by 1.09% and 1.05% because FCDH had more operations that used erosion and dilation to complement features than FCDL. After removing the cascade structure, the effect of the model was significantly degraded, demonstrating the superiority of

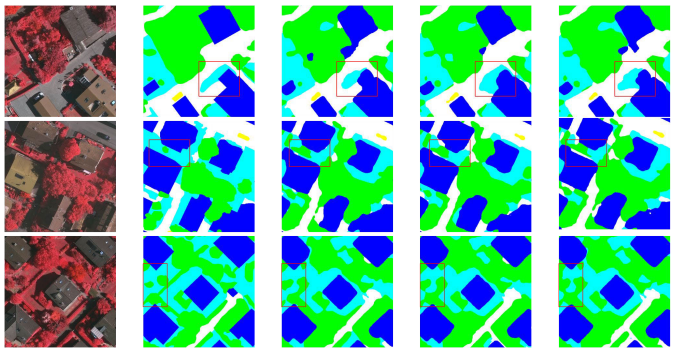


Fig. 10. Scene classification outcomes of validity of MCA and ECF: (a) IRRG, (b) label, (c) GTESNet-S*, (d) without MCA, and (e) without MCA and ECF.

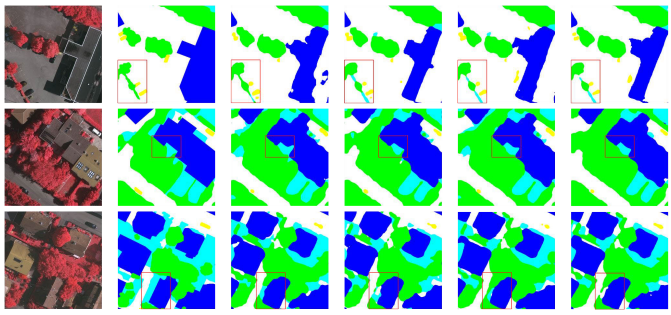


Fig. 11. Scene classification outcomes of validity of FCD: (a) IRRG, (b) label, (c) GTESNet-S*, (d) without FCDL, (e) without FCDH, and (f) without FCDH + FCDL.

using the output of our first-layer decoder to supervise and supplement the output of the second-layer decoder. This result is illustrated in Fig. 11.

V. Conclusion

We proposed the GTESNet-S* to improve the RSI scene classification performance. We developed a lightweight model that can observe the characteristics of different types of features, mine the contextual information relevant to the results, account for the quality differences among different modal images, and produce a high-quality prediction map. We implemented GTE to cluster features and mine contextual cues efficiently. The ECF and MCA generate modal quality weights and fuse the features, and the FCDs generate the prediction graphs. In addition, we introduced a KD scheme into the task to construct a lightweight model, allowing the GTESNet-S to learn three relatively high-quality knowledge items from two GTESNet-Ts, thereby improving the performance. Finally, we verified the feasibility of the GTESNet-T through extensive experiments using two datasets. In future studies, we will explore superior and even lighter models.

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